DATA ANALYTICS PROJECT

MCU — Commercial and Critical Performance Analysis Report

Analysis of Marvel Cinematic Universe Films — Phases 1 to 5

Period analyzed: 2008–2024

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Project **Data Analytics** Period **2008–2024**

Scope **Phases 1 to 5** Date **September 2026**

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1 Executive summary

1.1 Context

The purpose of this analysis is to study the commercial and critical performance of Marvel Cinematic Universe (MCU) films, in order to better understand the trends observed within the catalogue and the factors associated with film performance.

The analysis covers 34 films spanning MCU Phases 1 to 5, with information on release dates, production budgets, box-office revenue, calculated profit, return on investment (ROI), and critic and audience ratings.

1.2 Key findings

Across the dataset, the films represent a cumulative production budget of 6,947 million dollars and a cumulative worldwide box office of 31,129 million dollars.

The profit calculated according to the definition used in the dataset, namely:

$$\textbf{Profit} = \textbf{Worldwide box office} - \textbf{Production budget}$$

amounts to 24,182 million dollars.

The catalogue's average IMDb rating is 7.26/10.

The phase-level analysis highlights substantial differences. Phase 3 shows the highest average box office, average profit, and overall ROI in the dataset under study. Conversely, the average financial indicators observed in Phase 5 are weaker than those of the earlier phases.

The analysis of relationships between variables also shows a positive association between production budget and worldwide box office. A positive association is also observed between IMDb rating and worldwide box office.

These relationships must nevertheless be interpreted as statistical associations and not as causal relationships.

2 Objectives of the analysis

This is a personal project. To structure the analysis, it uses the following fictional business brief, written as a working framework (it is not an engagement carried out for a client):

> "We want to analyze the commercial and critical performance of Marvel Cinematic Universe films in order to better understand the factors associated with a film's success.
>
> In particular, we want to compare performance across MCU phases, examine the relationship between investment and revenue, analyze film profitability, and compare critical reception.
>
> From this analysis, we want indicators and visualizations that will help management better understand the catalogue's performance trends."

Based on this need, several analytical streams were selected:

- measure commercial performance;
- compare the different MCU phases;
- analyze the relationship between investment and revenue;
- measure profitability;
- compare critical performance;
- identify factors associated with commercial performance;
- observe performance trends over time.

3 Data and scope

3.1 Data source

The dataset used contains 34 MCU films covering the period 2008–2024 and includes 20 variables describing, in particular, temporal, commercial, financial, and critical information about the films.

The main sources mentioned in the dataset documentation are:

- IMDb;
- Box Office Mojo;
- Rotten Tomatoes;
- Wikipedia.

3.2 Main variables

The variables used in the analysis include, in particular:

- film title;
- release date and year;
- MCU phase;
- chronological order;
- director;
- lead actor;
- runtime;
- production budget;
- worldwide box office;
- domestic box office;
- profit;
- ROI;
- domestic box-office share;
- IMDb rating;
- Rotten Tomatoes score;
- Metacritic score.

The dataset defines profit as the difference between worldwide box office and production budget, while ROI corresponds to the return calculated from profit and production budget.

4 Data preparation and quality

4.1 Initial profiling

A technical profiling of the dataset was carried out before the analysis.

The checks confirmed:

- 34 rows and 20 columns;
- no missing values;
- no duplicates;
- 34 unique film identifiers;
- no inconsistency between phase and `phase_number`;
- no inconsistency between `release_date` and `release_year`;
- no critical values outside the expected ranges;
- no negative runtime.

4.2 Consistency checks

Additional checks were performed on the calculated variables.

The values of:

- `profit_millions`;
- `domestic_percentage`;
- `roi_percentage`

were compared with their respective calculation formulas.

No inconsistency was detected.

4.3 Cleaning

The main correction identified concerned the data type of `release_date`, which was initially stored as text.

The column was converted to a true date type before the cleaned dataset was exported.

The cleaned file was then used for modeling and analysis in Power BI.

Quality conclusion: the checks performed indicate that the dataset is consistent and suitable for the selected analysis, with a data-type correction applied to the release date.

5 Methodology and analytical architecture

The analysis was carried out in several stages:

Step	Stage	Description
1	Raw data	Starting point: MCU dataset
2	Profiling	Analysis of structure, types, missing values, duplicates, and distributions
3	Quality control	Verification of data validity, consistency, and integrity
4	Cleaning	Type correction and preparation of a usable file
5	Exploratory analysis	Data exploration with Python/Pandas and identification of the main findings
6	Modeling	Construction of the star schema in Power BI
7	Indicators	Creation of analytical measures with DAX
8	Visualization and analysis	Construction of the dashboard and interpretation of the results

The Power BI model is based on a star schema comprising:

- `Fact_Movie`;
- `Dim_Movie`;
- `Dim_Phase`;
- `Dim_Date`.

The granularity of the fact table is:

1 film = 1 row in `Fact_Movie`.

This structure makes it possible, in particular, to analyze performance by film, phase, and release period.

6 Results analysis

6.1 Catalogue overview

The main catalogue indicators are as follows:

Indicator	Result
Number of films	34
Total budget	6,947 M$
Total worldwide box office	31,129 M$
Total calculated profit	24,182 M$
Average worldwide box office	915.56 M$
Average budget	204.32 M$
Average ROI per film	339.58 %
Average IMDb rating	7.26/10
Average Rotten Tomatoes score	80.71 %
Average Metacritic score	66/100

The dashboard screenshot below was captured with Phase 3 selected: the KPI cards therefore show the values for that phase (11 films, 13,500 M$ worldwide box office, 11,179 M$ profit, average IMDb rating of 7.59), not those of the whole catalogue, which are given in the table above.

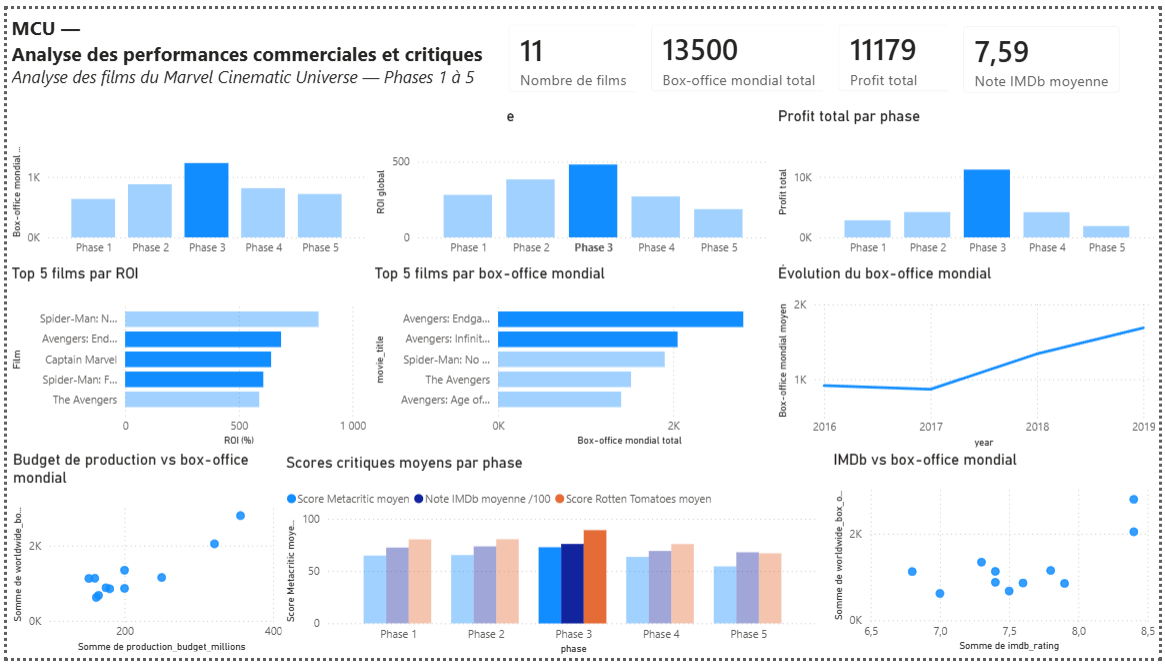


Figure 1: MCU Power BI dashboard (Phase 3 selected)

6.2 Commercial performance by phase

Average worldwide box office varies substantially across phases.

The observed averages are:

Phase	Average worldwide box office
Phase 1	634.83 M$
Phase 2	877.67 M$
Phase 3	1,227.27 M$
Phase 4	812.71 M$
Phase 5	716.25 M$

Phase 3 therefore shows the highest average worldwide box office within the scope under study.

6.3 Highest-grossing films

The five films with the highest worldwide box office in the dataset are:

Film	Worldwide box office
Avengers: Endgame	2,798 M$
Avengers: Infinity War	2,048 M$
Spider-Man: No Way Home	1,902 M$
The Avengers	1,518 M$
Avengers: Age of Ultron	1,405 M$

These results show that a small number of films contribute strongly to the catalogue's overall commercial performance.

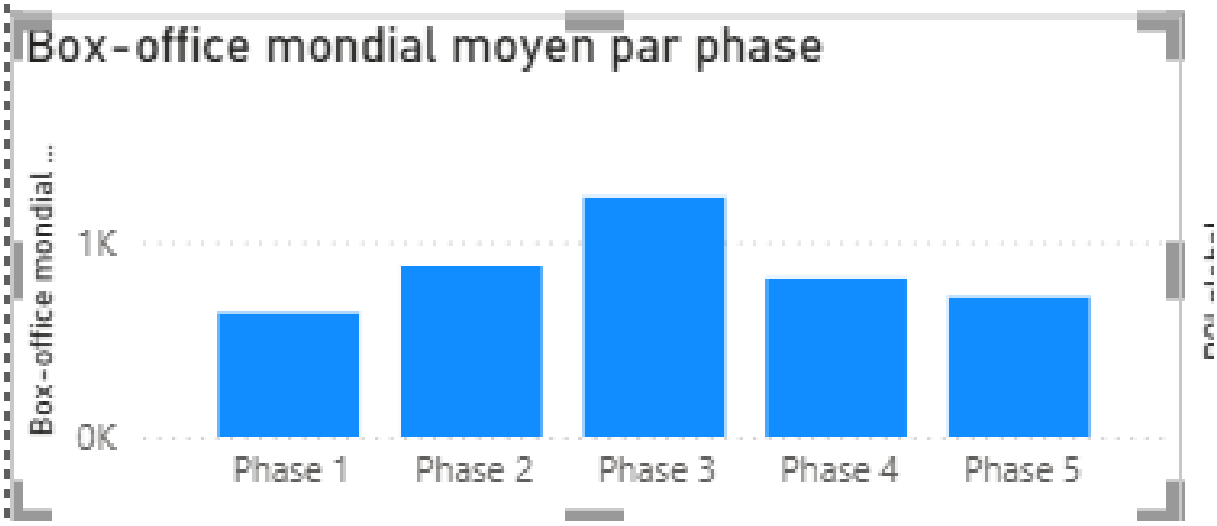


Figure 2: Average worldwide box office by phase

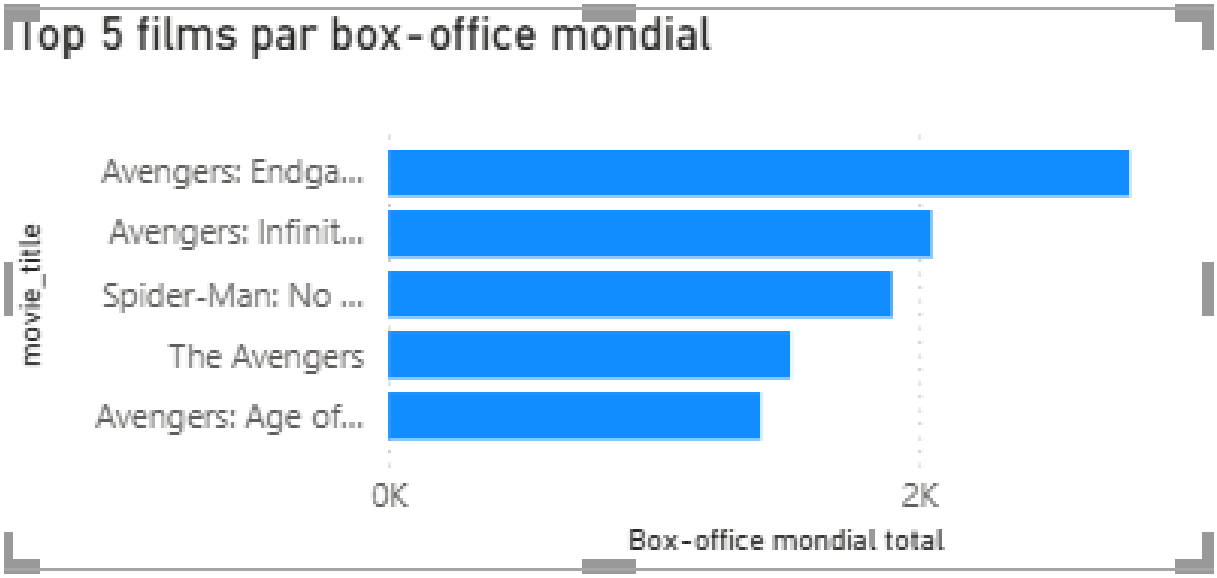


Figure 3: Top 5 films by worldwide box office

6.4 Box-office trends over time

The annual trend shows substantial variation in average commercial performance over the period under study.

The highest annual averages are those of 2019 (1,686 M$), 2012 (1,518 M$), 2018 (1,338.67 M$) and 2024 (1,338 M$). Each year includes only one to four films, and no film was released in 2020, so these annual averages depend heavily on a few titles.

The trend is therefore not linear: several periods of growth and decline are observed.

6.5 Profitability and return on investment

Overall ROI was calculated using the formula:

$$\textbf{Overall ROI} = \textbf{Total profit} \ / \ \textbf{Total budget} \times \textbf{100}$$

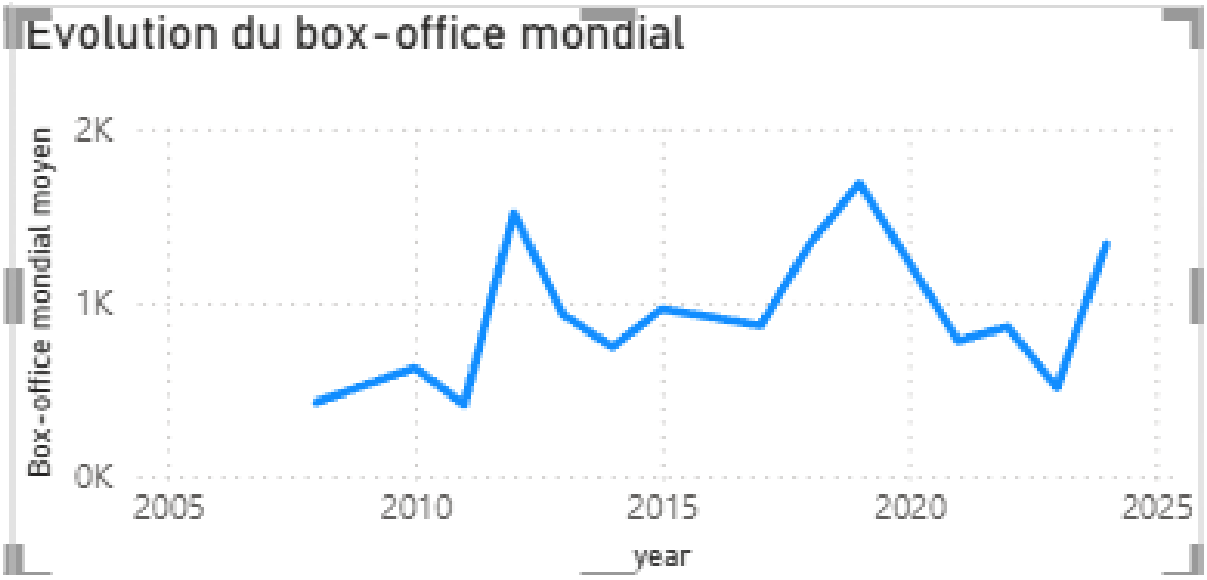


Figure 4: Worldwide box office over time

This method measures the overall efficiency of investments rather than simply averaging individual ROI values.

The results by phase are:

Phase	**Overall ROI**
Phase 1	280.90 %
Phase 2	383.12 %
Phase 3	481.65 %
Phase 4	270.38 %
Phase 5	186.50 %

Across the whole catalogue, overall ROI is 348.09 % (24,182 / 6,947), compared with 339.58 % for the average of individual ROI values.

Phase 3 therefore has the highest overall ROI among the five phases studied.

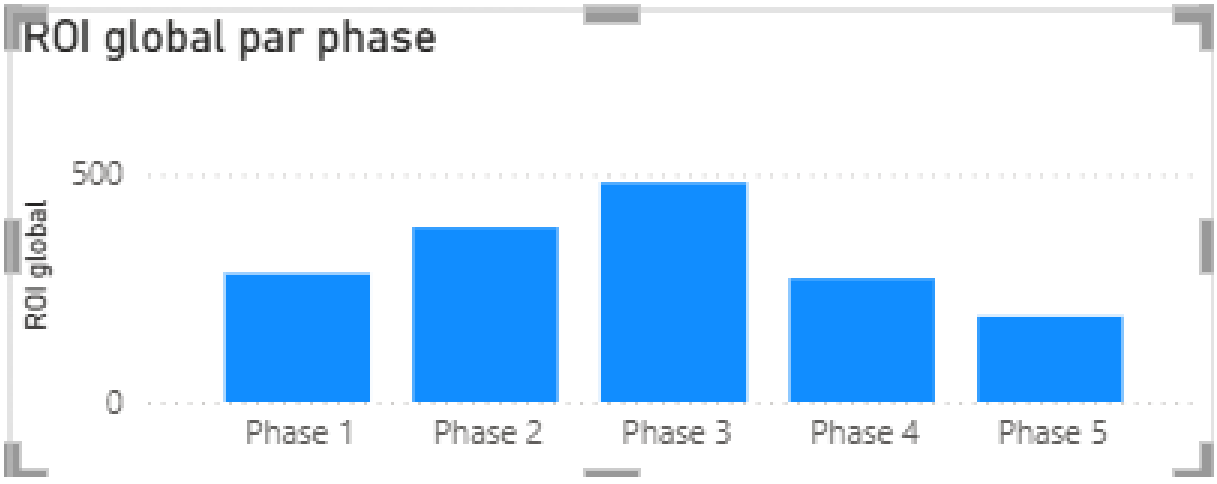


Figure 5: Overall ROI by phase

6.6 Relationship between budget and box office

The analysis of production budget and worldwide box office shows a positive correlation of approximately 0.63 in the dataset.

This indicates that, within the data under study, films with higher budgets also tend to be associated with higher box-office results.

However, this relationship does not establish that increasing the budget automatically causes an increase in box office.

Other factors may be involved and are not directly measured in the dataset.

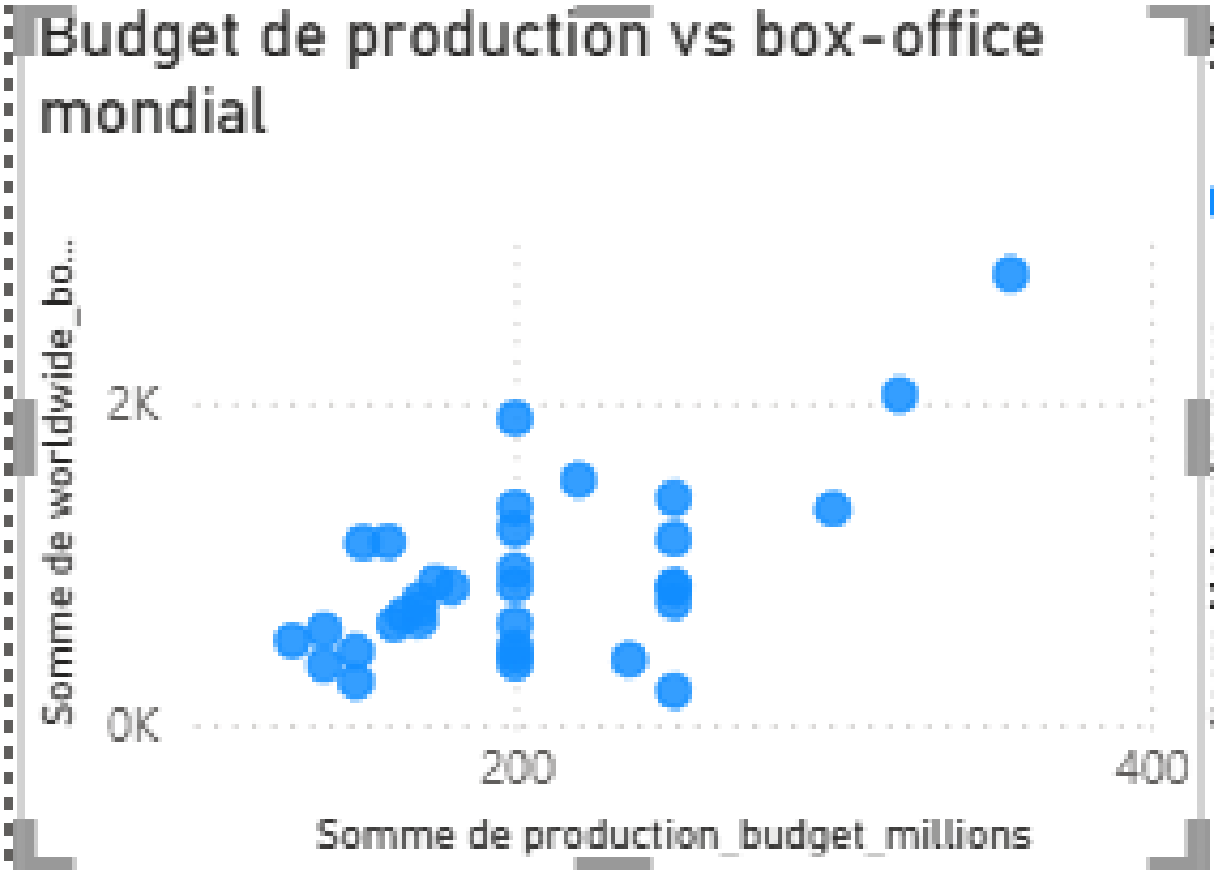


Figure 6: Production budget vs worldwide box office

6.7 Films with the highest ROI

The analysis of individual ROI identifies the films that generated the highest return relative to their production budget.

The main results include, in particular:

- Spider-Man: No Way Home;
- Avengers: Endgame;
- Captain Marvel;
- Spider-Man: Far From Home;
- The Avengers.

ROI is used here as a complementary indicator to box office: a film generating lower revenue

may show a higher return if it was produced with a relatively smaller investment.

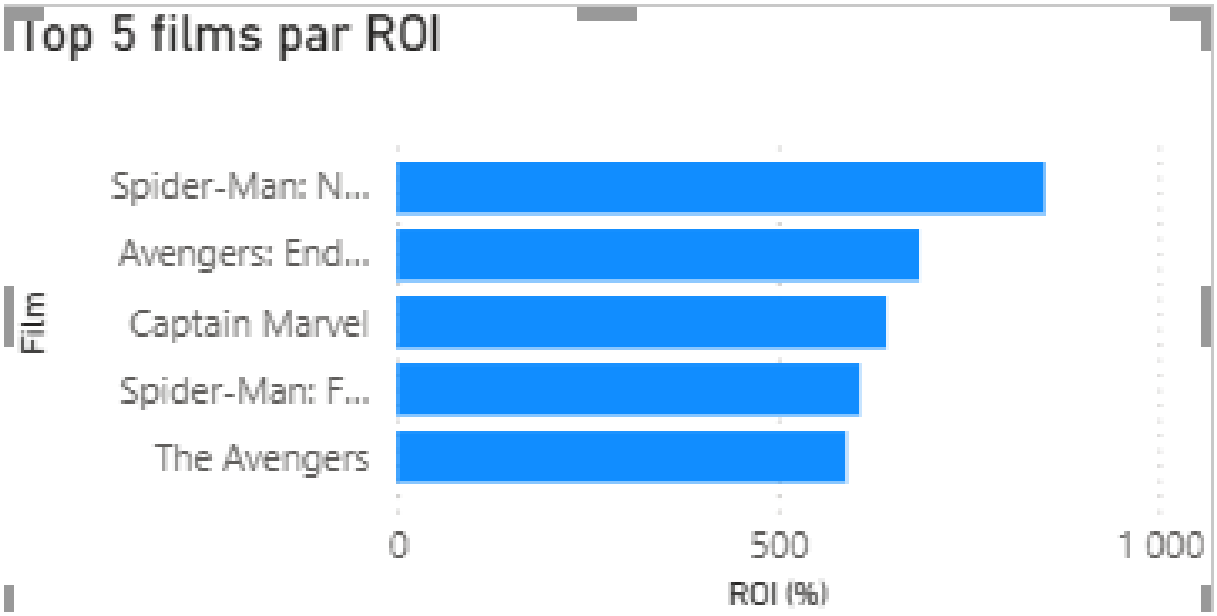


Figure 7: Top 5 films by ROI

6.8 Profit by phase

Total profit also varies substantially across phases.

The results show a particularly large contribution from Phase 3, which generates the highest total profit in the dataset.

This observation is consistent with its high box-office level and overall ROI.

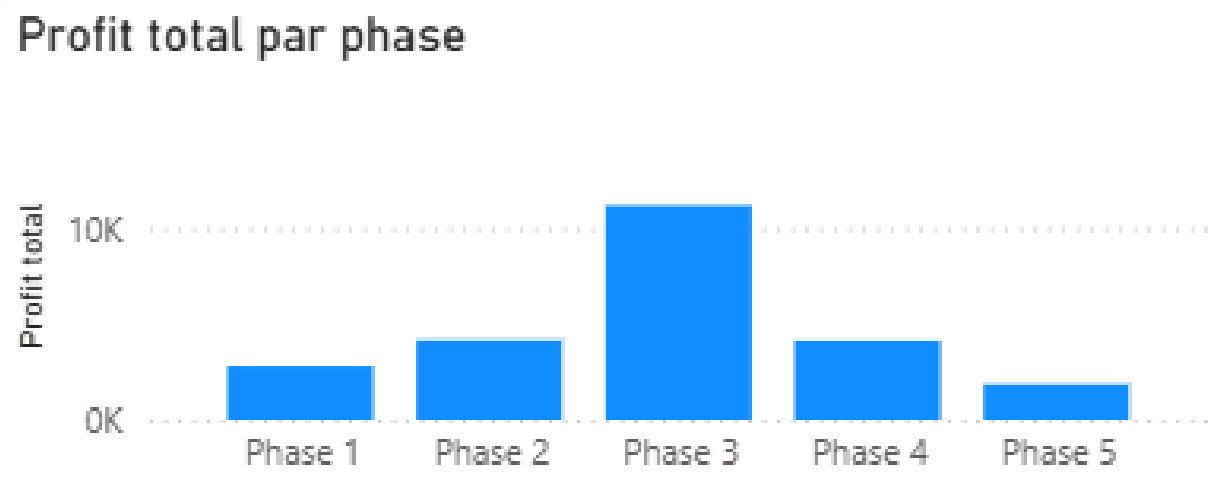


Figure 8: Total profit by phase

6.9 Critical performance

Three critical indicators were compared:

- IMDb;
- Rotten Tomatoes;
- Metacritic.

To facilitate visual comparison, the IMDb rating was converted to a 100-point scale for this visualization only.

The observed averages by phase are:

Phase	IMDb /100	Rotten Tomatoes	Metacritic
Phase 1	72.5	80.33	64.83
Phase 2	73.7	80.50	65.33
Phase 3	75.9	89.09	72.73
Phase 4	69.3	75.86	63.57
Phase 5	68.0	67.00	54.50

In the dataset under study, Phase 3 has the highest average critic scores for all three indicators.

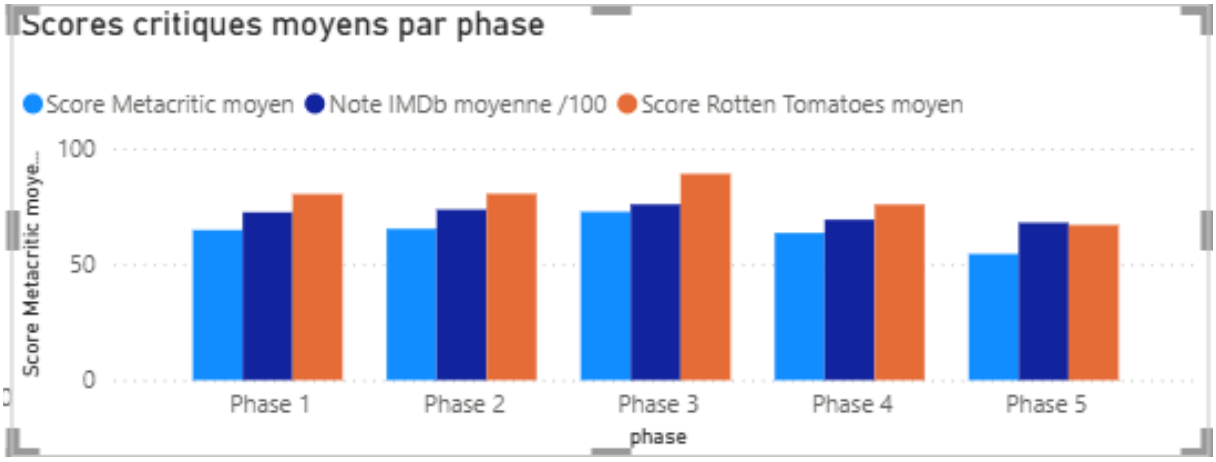


Figure 9: Average critic scores by phase

6.10 Critical reception and box office

The exploratory analysis shows a correlation of approximately 0.66 between IMDb rating and worldwide box office.

This relationship is stronger than those observed with Rotten Tomatoes and Metacritic:

- IMDb ↔ Box office: 0.66
- Rotten Tomatoes ↔ Box office: 0.44
- Metacritic ↔ Box office: 0.42

In this dataset, IMDb rating therefore shows the strongest statistical association with box office among the three critic indicators studied.

It is nevertheless important to recall that a correlation does not demonstrate a causal relationship.

7 Conclusions

The analysis of the 34 films under study highlights several findings.

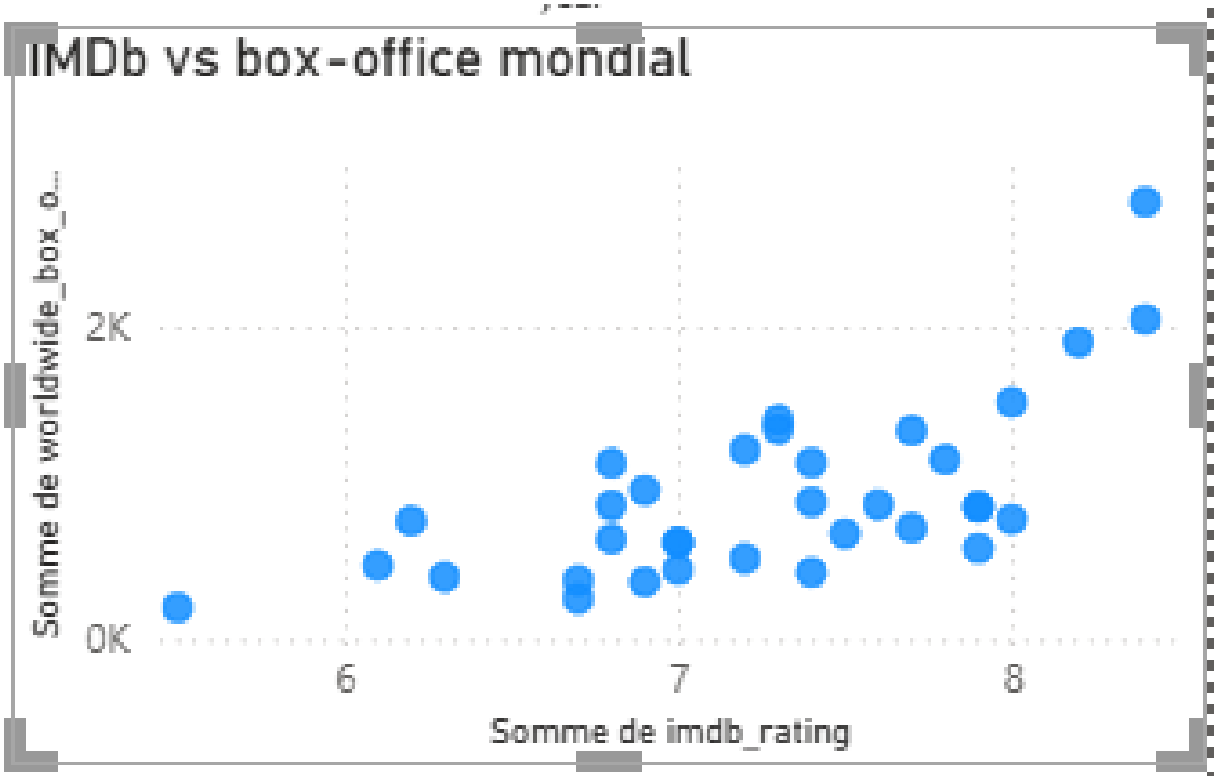


Figure 10: IMDb vs worldwide box office

First, commercial and financial performance varies substantially across MCU phases. Phase 3 shows the highest average box-office levels as well as the highest overall ROI and total profit within the scope under study.

Second, a small number of films account for a large share of the catalogue's commercial performance. Avengers: Endgame, Avengers: Infinity War, and Spider-Man: No Way Home are among the films generating the highest worldwide box-office results.

Third, the analysis highlights a positive association between production budget and worldwide box office. However, budget alone is not sufficient to explain commercial performance.

Fourth, critical reception also shows a positive association with box office, particularly in the case of IMDb rating. This observation suggests that it may be relevant to study commercial and critical indicators together.

Finally, the results show a decline in average indicators of commercial performance, profitability, and critical reception between some recent phases and Phase 3. This trend is a point of attention that would warrant further analysis.

8 Recommendations

The following recommendations are based directly on the observed results.

8.1 Monitor commercial performance and profitability together

Box office alone cannot measure the efficiency of an investment.

It is therefore recommended to track simultaneously:

- budget;
- box office;
- profit;
- ROI.

This approach makes it possible to distinguish films that generate high revenue from those that convert their investment into profit efficiently.

8.2 Compare performance across phases systematically

The gaps observed between phases show the value of periodic monitoring of indicators by period.

A dashboard comparing each new phase with previous phases could help identify performance changes quickly.

8.3 Examine factors associated with commercial performance

The observed relationship between budget and box office, as well as that between IMDb and box office, justifies further analysis of the factors associated with commercial success.

A future analysis could include additional variables such as:

- genre;
- lead character;
- director;
- franchise;
- release calendar;
- marketing;
- competition at the time of release;
- economic context;
- theatrical run length.

This would make it possible to go beyond the variables currently available.

8.4 Pay particular attention to the performance of recent phases

The average indicators observed for Phases 4 and 5 are lower than those of Phase 3 on several dimensions.

It would therefore be relevant to carry out a more detailed analysis of the films concerned in order to identify characteristics common to the weakest and strongest performances.

This analysis should distinguish general trends from individual cases, in order to avoid drawing conclusions from a limited number of films.

8.5 Establish regular indicator monitoring

The Power BI dashboard can serve as the basis for a performance monitoring system.

The main indicators to track would include:

Commercial performance

- worldwide box office;
- average box office;

- trend over time.

Profitability

- profit;
- ROI;
- budget.

Critical performance

- IMDb;
- Rotten Tomatoes;
- Metacritic.

Regular use of these indicators would make it possible to track the catalogue's evolution and compare new releases with historical performance.

9 Limitations of the analysis

Several limitations must be taken into account when interpreting the results.

9.1 Dataset size

The analysis covers only 34 films. The results must therefore be interpreted within the scope of this dataset and should not be automatically generalized to the film industry as a whole.

9.2 Definition of profit

The profit used in this analysis corresponds to:

Worldwide box office − production budget

It is therefore a calculated indicator, and not necessarily a studio's actual accounting profit.

Marketing, distribution, taxes, commissions, and other expenses are not included in this formula.

9.3 Correlation and causality

The observed correlations make it possible to identify associations between variables, but they do not establish causality.

For example, a positive correlation between budget and box office does not necessarily mean that a higher budget directly generates higher revenue.

9.4 Unavailable variables

Some factors that may influence commercial performance are not included in the dataset.

The results should therefore be regarded as a descriptive and exploratory analysis of the catalogue under study.

10 Reporting tool

An interactive Power BI dashboard was developed to support exploration of the results.

The dashboard brings together:

- the main KPIs;
- performance by phase;
- the highest-performing films;
- the trend over time;
- ROI;
- profit;
- critic scores;
- the budget/box-office relationship;
- the IMDb/box-office relationship.

The visualizations were selected on the basis of the business needs, in order to avoid multiplying charts without additional analytical value. The final selection includes 9 main visualizations covering the commercial, financial, critical, and temporal dimensions of the problem.

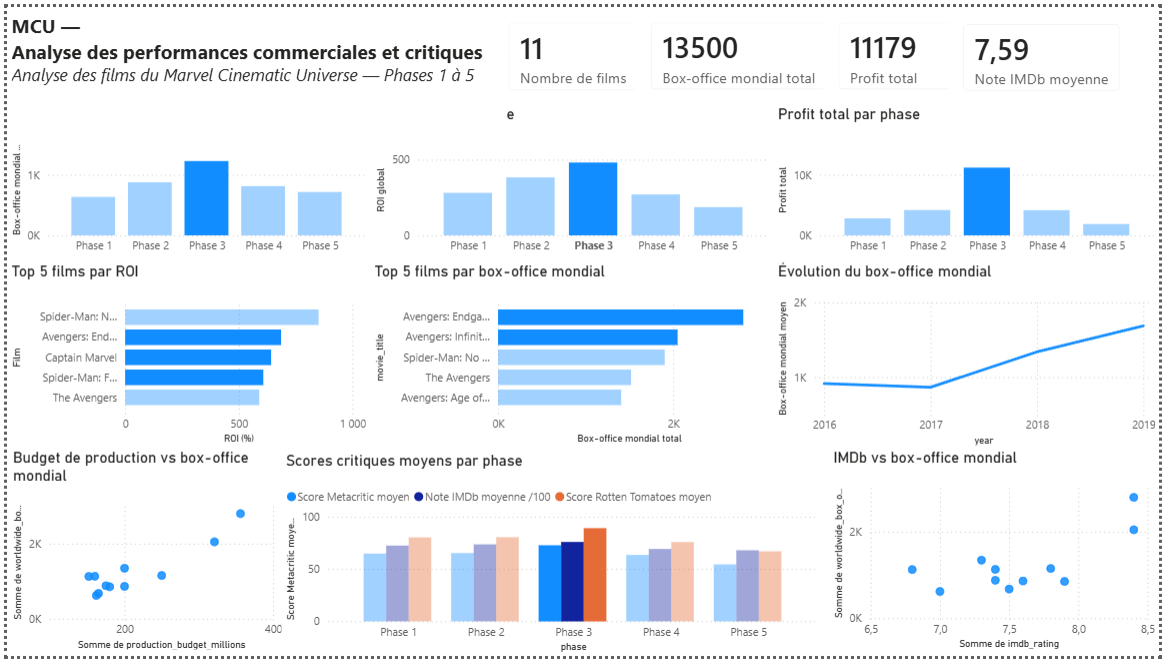


Figure 11: Final Power BI dashboard (Phase 3 selected)

11 Final conclusion

This analysis made it possible to build a complete chain from understanding the business need through to interactive reporting of the results.

The project covered:

- problem framing;
- data inventory and understanding;
- profiling;
- quality control;

- cleaning;
- exploratory analysis with Python;
- dimensional modeling;
- creation of DAX measures;
- design of visualizations aligned with the business questions;
- delivery of an interactive Power BI dashboard.

The results provide a clearer understanding of performance differences across MCU phases, the relationship between investment and revenue, film profitability, and the associations between critical reception and commercial performance.

The dashboard is therefore a synthesis tool for exploring catalogue performance, while this report provides the context, methods, main findings, conclusions, and avenues for further analysis.

12 Appendices

12.1 Appendix A — List of visualizations

No.	Visualization
1	Average worldwide box office by phase
2	Top 5 films by worldwide box office
3	Worldwide box office over time
4	Overall ROI by phase
5	Production budget vs worldwide box office
6	Top 5 films by ROI
7	Total profit by phase
8	Average critic scores by phase
9	IMDb vs worldwide box office

This selection corresponds to the final design defined for the dashboard.

12.2 Appendix B — Project files

The main technical files of the project include, in particular:

- original dataset;
- cleaned dataset;
- Python scripts for profiling, cleaning, and analysis;
- Power BI model;
- analysis report.